EXPERIMENT 1 — FIND-S ALGORITHM

Aim : To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Algorithm / Procedure :

1. Define the training dataset.
2. Initialize the hypothesis with the most specific values.
3. Consider only positive training examples.
4. Compare positive examples with the current hypothesis.
5. Replace differing attributes with ?.
6. Ignore negative examples.
7. Display the final hypothesis.

Dataset 1: Play Tennis

Sky	AirTemp	Humidity	Wind	Water	Forecast	PlayTennis
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Same	Yes
Rainy	Warm	Normal	Weak	Warm	Same	No
Sunny	Warm	Normal	Weak	Warm	Same	Yes

Target Attribute: PlayTennis

Code :

```
# FIND-S Algorithm - Dataset 1: Play Tennis

data = [

    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
    ["Sunny", "Warm", "High", "Strong", "Cool", "Same", "Yes"],
    ["Rainy", "Warm", "Normal", "Weak", "Warm", "Same", "No"],
    ["Sunny", "Warm", "Normal", "Weak", "Warm", "Same", "Yes"]
```

```
]
hypothesis = ["Ø", "Ø", "Ø", "Ø", "Ø", "Ø"]

for example in data:
    if example[-1] == "Yes":
        for i in range(6):
            if hypothesis[i] == "Ø":
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = "?"

print("Final Most Specific Hypothesis:")
print(hypothesis)
```

Output :

```
------------------------------------------------------------
DATASET 1: PLAY TENNIS
------------------------------------------------------------
Attributes: ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast']
Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', '?', '?', 'Same']
```

Dataset 2: Loan Approval

Income	CreditScore	Employment	Property	Age	LoanApproved
High	Good	Permanent	Yes	Young	Yes
High	Good	Permanent	No	Middle	Yes
Low	Poor	Temporary	No	Young	No
Medium	Good	Permanent	Yes	Middle	Yes
High	Average	Temporary	Yes	Old	No
High	Good	Permanent	Yes	Middle	Yes

Target Attribute: LoanApproved

Code :

```
# FIND-S Algorithm - Dataset 2: Loan Approval
data = [
    ["High", "Good", "Permanent", "Yes", "Young", "Yes"],
```

```
    ["High", "Good", "Permanent", "No", "Middle", "Yes"],
    ["Low", "Poor", "Temporary", "No", "Young", "No"],
    ["Medium", "Good", "Permanent", "Yes", "Middle", "Yes"],
    ["High", "Average", "Temporary", "Yes", "Old", "No"],
    ["High", "Good", "Permanent", "Yes", "Middle", "Yes"]
]
hypothesis = ["Ø", "Ø", "Ø", "Ø", "Ø"]
for example in data:
    if example[-1] == "Yes":
        for i in range(5):
            if hypothesis[i] == "Ø":
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = "?"

print("Final Most Specific Hypothesis:")
print(hypothesis)
```

Output :

```
------------------------------------------------------------
DATASET 2: LOAN APPROVAL
------------------------------------------------------------
Attributes: ['Income', 'CreditScore', 'Employment', 'Property', 'Age']
Final Most Specific Hypothesis:
['?', 'Good', 'Permanent', '?', '?']
```

Dataset 3: Student Placement

CGPA	Communication	Internship	Programming	Aptitude	Placed
High	Good	Yes	Good	High	Yes
High	Excellent	Yes	Good	High	Yes
Medium	Average	No	Average	Medium	No
High	Good	Yes	Excellent	High	Yes
Low	Poor	No	Average	Low	No
High	Good	Yes	Good	Medium	Yes

Target Attribute: Placed

Code :

```
# FIND-S Algorithm - Dataset 3: Student Placement

data = [
    ["High", "Good", "Yes", "Good", "High", "Yes"],
    ["High", "Excellent", "Yes", "Good", "High", "Yes"],
    ["Medium", "Average", "No", "Average", "Medium", "No"],
    ["High", "Good", "Yes", "Excellent", "High", "Yes"],
    ["Low", "Poor", "No", "Average", "Low", "No"],
    ["High", "Good", "Yes", "Good", "Medium", "Yes"]
]

hypothesis = ["Ø", "Ø", "Ø", "Ø", "Ø"]

for example in data:
    if example[-1] == "Yes":
        for
 i in range(5):
            if hypothesis[i] == "Ø":
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = "?"

print("Final Most Specific Hypothesis:")
print(hypothesis)
```

Output :

```
------------------------------------------------------------
DATASET 3: STUDENT PLACEMENT
------------------------------------------------------------
Attributes: ['CGPA', 'Communication', 'Internship', 'Programming', 'Aptitude']
Final Most Specific Hypothesis:
['High', '?', 'Yes', '?', '?']
```

Dataset 4: Disease Diagnosis

Fever	Cough	Headache	BodyPain	Fatigue	Disease
Yes	Yes	Yes	Yes	Yes	Positive
Yes	Yes	No	Yes	Yes	Positive
No	Yes	Yes	No	No	Negative
Yes	Yes	Yes	No	Yes	Positive
No	No	Yes	Yes	No	Negative
Yes	Yes	No	No	Yes	Positive

Target Attribute: Disease

Code :

```
# FIND-S Algorithm - Dataset 4: Disease Diagnosis

data = [
    ["Yes", "Yes", "Yes", "Yes", "Yes", "Positive"],
    ["Yes", "Yes", "No", "Yes", "Yes", "Positive"],
    ["No", "Yes", "Yes", "No", "No", "Negative"],
    ["Yes", "Yes", "Yes", "No", "Yes", "Positive"],
    ["No", "No", "Yes", "Yes", "No", "Negative"],
    ["Yes", "Yes", "No", "No", "Yes", "Positive"]
]

hypothesis = ["Ø", "Ø", "Ø", "Ø", "Ø"]

for example in data:
    if example[-1] == "Positive":
        for i in range(5):
            if hypothesis[i] == "Ø":
                hypothesis[i] = example[i]
            elif hypothesis[i] != example[i]:
                hypothesis[i] = "?"

print("Final Most Specific Hypothesis:")
print(hypothesis)
```

Output :

```
------------------------------------------------------------
DATASET 4: DISEASE DIAGNOSIS
------------------------------------------------------------
Attributes: ['Fever', 'Cough', 'Headache', 'BodyPain', 'Fatigue']
Final Most Specific Hypothesis:
['Yes', 'Yes', '?', '?', 'Yes']
```

Dataset 5: Email Spam Detection

ContainsLink	ContainsOffer	UnknownSender	Attachment	ManyRecipients	Spam
Yes	Yes	Yes	No	Yes	Yes
Yes	Yes	Yes	Yes	Yes	Yes
No	No	No	No	No	No
Yes	No	Yes	No	Yes	Yes
No	Yes	No	Yes	No	No
Yes	Yes	Yes	No	No	Yes

Target Attribute: Spam

Code :

```
# FIND-S Algorithm - Dataset 5: Email Spam Detection

data = [
    ["Yes", "Yes", "Yes", "No", "Yes", "Yes"],
    ["Yes", "Yes", "Yes", "Yes", "Yes", "Yes"],
    ["No", "No", "No", "No", "No", "No"],
    ["Yes", "No", "Yes", "No", "Yes", "Yes"],
    ["No", "Yes", "No", "Yes", "No", "No"],
    ["Yes", "Yes", "Yes", "No", "No", "Yes"]
]

hypothesis = ["Ø", "Ø", "Ø", "Ø", "Ø"]

for example in data:
    if example[-1] == "Yes":
        for i in range(5):
            if hypothesis[i] == "Ø":
```

```
            hypothesis[i] = example[i]
        elif hypothesis[i] != example[i]:
            hypothesis[i] = "?"

print("Final Most Specific Hypothesis:")
print(hypothesis)
```

Output :

```
------------------------------------------------------------
DATASET 5: EMAIL SPAM DETECTION
------------------------------------------------------------
Attributes: ['ContainsLink', 'ContainsOffer', 'UnknownSender', 'Attachment', 'ManyRecipients']
Final Most Specific Hypothesis:
['Yes', '?', 'Yes', '?', '?']
```

Result :

Thus, the FIND-S algorithm was successfully implemented in Python and the most specific hypothesis was obtained for all five given training datasets.